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MIDTERM ASSEMBLY LANGUAGE

SOURCE CODE

global _start

section .data

; ANSI cursor-control sequences

clear_screen db 27, "[2J", 27, "[3J", 27, "[H"

clear_screen_len equ \$ - clear_screen

cursor_prefix db 27, "["

cursor_prefix_len equ \$ - cursor_prefix

semicolon db ";

semicolon_len equ \$ - semicolon

cursor_home db "H"

cursor_home_len equ \$ - cursor_home

newline db 10

; Page positioning constants

PAGE_RIGHT equ 72

LEFT_BOX_RIGHT equ 36

RIGHT_BOX_LEFT equ 37

HALF_BOX_WIDTH equ 36

; -----

; Questions

; -----

input_title db "ABOUT ME SLAMBOOK - ENTER YOUR INFORMATION"

input_title_len equ \$ - input_title

input_instruction db "Type an answer after each question and press ENTER."

input_instruction_len equ \$ - input_instruction

p_name db "Name: "

p_name_len equ \$ - p_name

p_email db "School Email: "

p_email_len equ \$ - p_email

p_course db "Course: "
p_course_len equ \$ - p_course

p_age db "Age: "
p_age_len equ \$ - p_age

p_happy db "First time I felt completely happy: "
p_happy_len equ \$ - p_happy

p_achievement db "First big achievement: "
p_achievement_len equ \$ - p_achievement

p_risk db "First risk I ever took: "
p_risk_len equ \$ - p_risk

p_colors db "Favorite colors: "
p_colors_len equ \$ - p_colors

p_song db "Favorite song: "
p_song_len equ \$ - p_song

p_singer db "Favorite singer: "
p_singer_len equ \$ - p_singer

p_food db "Favorite food: "
p_food_len equ \$ - p_food

p_weekend db "Favorite thing to do on a weekend: "
p_weekend_len equ \$ - p_weekend

p_hobbies db "Hobbies: "
p_hobbies_len equ \$ - p_hobbies

p_ambition db "Ambition: "
p_ambition_len equ \$ - p_ambition

p_motto db "Motto: "
p_motto_len equ \$ - p_motto

ready_message db "Your answers are ready. Press ENTER to view your
slambook."
ready_message_len equ \$ - ready_message

```

; -----
; Display labels
; -----

about_title db "[ ABOUT ME ]"
about_title_len equ $ - about_title

name_label db "Name:"
name_label_len equ $ - name_label

email_label db "School Email:"
email_label_len equ $ - email_label

course_label db "Course:"
course_label_len equ $ - course_label

age_label db "Age:"
age_label_len equ $ - age_label

firsts_title db "MY FIRSTS"
firsts_title_len equ $ - firsts_title

happy_label db "First time I felt completely happy:"
happy_label_len equ $ - happy_label

achievement_label db "First big achievement:"
achievement_label_len equ $ - achievement_label

risk_label db "First risk I ever took:"
risk_label_len equ $ - risk_label

favorites_title db "FAVORITES"
favorites_title_len equ $ - favorites_title

colors_label db "Colors:"
colors_label_len equ $ - colors_label

song_label db "Song:"
song_label_len equ $ - song_label

singer_label db "Singer:"
singer_label_len equ $ - singer_label

food_label db "Food:"

```

food_label_len equ \$ - food_label

weekend_label db "Weekend:"

weekend_label_len equ \$ - weekend_label

hobbies_label db "Hobbies:"

hobbies_label_len equ \$ - hobbies_label

ambition_title db "AMBITION"

ambition_title_len equ \$ - ambition_title

ambition_label db "Ambition:"

ambition_label_len equ \$ - ambition_label

motto_label db "Motto:"

motto_label_len equ \$ - motto_label

footer db "***** MY SLAMBOOK *****"

footer_len equ \$ - footer

; -----

; ASCII art header

; -----

header1 db "AAAAA BBBB OOO U U TTTT M M EEEEE"

header1_len equ \$ - header1

header2 db " A A B B O O U U T MM MM E"

header2_len equ \$ - header2

header3 db "AAAAA BBBB O O U U T M M M EEEE"

header3_len equ \$ - header3

header4 db " A A B B O O U U T M M E"

header4_len equ \$ - header4

header5 db " A A BBBB OOO UUUUU T M M EEEEE"

header5_len equ \$ - header5

cat1 db " ^\ _ ^\ "

cat1_len equ \$ - cat1

cat2 db "(o o)"

cat2_len equ \$ - cat2

course_buf times 80 db 0
age_buf times 16 db 0

happy_buf times 100 db 0
achievement_buf times 100 db 0
risk_buf times 100 db 0

colors_buf times 80 db 0
song_buf times 80 db 0
singer_buf times 80 db 0
food_buf times 80 db 0
weekend_buf times 100 db 0
hobbies_buf times 100 db 0

ambition_buf times 100 db 0
motto_buf times 100 db 0
wait_buf times 8 db 0

number_buffer times 12 db 0
hline_buffer times 100 db 0
one_char db 0

section .text

```
;-----  
; print_at(row, column, address, length)  
;-----
```

print_at:

push ebp
mov ebp, esp
push ebx

push dword [ebp + 12]
push dword [ebp + 8]
call goto_xy
add esp, 8

mov eax, 4
mov ebx, 1
mov ecx, [ebp + 16]
mov edx, [ebp + 20]
int 0x80

```
pop ebx
leave
ret
```

```
; -----
; print_cstr_at(row, column, address)
; -----
```

```
print_cstr_at:
    push ebp
    mov ebp, esp
    push ebx

    mov eax, [ebp + 16]
    call strlen

    push eax
    push dword [ebp + 16]
    push dword [ebp + 12]
    push dword [ebp + 8]
    call print_at
    add esp, 16

    pop ebx
    leave
    ret
```

```
; -----
; print_cstr_n_at(row, column, address, maximum_length)
; Prevents long input from overwriting borders.
; -----
```

```
print_cstr_n_at:
    push ebp
    mov ebp, esp
    push ebx
    push esi
    push edi

    mov esi, [ebp + 16]
    mov edi, [ebp + 20]
```

```

    xor ecx, ecx

.count:
    cmp ecx, edi
    jae .print

    cmp byte [esi + ecx], 0
    je .print

    inc ecx
    jmp .count

.print:
    push ecx
    push dword [ebp + 16]
    push dword [ebp + 12]
    push dword [ebp + 8]
    call print_at
    add esp, 16

    pop edi
    pop esi
    pop ebx
    leave
    ret

; -----
; strlen
; -----

strlen:
    push edi
    mov edi, eax
    xor eax, eax

.count:
    cmp byte [edi + eax], 0
    je .finished
    inc eax
    jmp .count

.finished:
    pop edi

```

ret

```
;-----  
; read_input(buffer, maximum_bytes)  
;-----
```

read_input:

push ebp
mov ebp, esp
push ebx
push esi

mov eax, 3
mov ebx, 0
mov ecx, [ebp + 8]
mov edx, [ebp + 12]
int 0x80

mov ecx, eax
mov esi, [ebp + 8]

test ecx, ecx
jz .empty

.scan:

cmp byte [esi], 10
je .terminate

cmp byte [esi], 13
je .terminate

inc esi
loop .scan

mov byte [esi], 0
jmp .done

.terminate:

mov byte [esi], 0
jmp .done

.empty:

mov byte [esi], 0

```
.done:
    pop esi
    pop ebx
    leave
    ret
```

```
; -----
; goto_xy(row, column)
; -----
```

```
goto_xy:
    push ebp
    mov ebp, esp
    push ebx

    mov eax, 4
    mov ebx, 1
    mov ecx, cursor_prefix
    mov edx, cursor_prefix_len
    int 0x80

    mov eax, [ebp + 8]
    call print_uint

    mov eax, 4
    mov ebx, 1
    mov ecx, semicolon
    mov edx, semicolon_len
    int 0x80

    mov eax, [ebp + 12]
    call print_uint

    mov eax, 4
    mov ebx, 1
    mov ecx, cursor_home
    mov edx, cursor_home_len
    int 0x80

    pop ebx
    leave
    ret
```

```
; -----  
; print_uint  
; -----
```

print_uint:

```
    push ebx  
    push ecx  
    push edx  
    push esi
```

```
    mov esi, number_buffer + 12  
    mov ebx, 10  
    xor ecx, ecx
```

```
    test eax, eax  
    jnz .convert
```

```
    dec esi  
    mov byte [esi], '0'  
    inc ecx  
    jmp .write
```

.convert:

```
    xor edx, edx  
    div ebx  
    add dl, '0'  
    dec esi  
    mov [esi], dl  
    inc ecx  
    test eax, eax  
    jnz .convert
```

.write:

```
    mov eax, 4  
    mov ebx, 1  
    mov edx, ecx  
    mov ecx, esi  
    int 0x80
```

```
    pop esi  
    pop edx  
    pop ecx
```

```
pop ebx
ret
```

```
; -----
; draw_hline(row, column, count, character)
; -----
```

```
draw_hline:
```

```
push ebp
mov ebp, esp
push ebx
push edi
```

```
push dword [ebp + 12]
push dword [ebp + 8]
call goto_xy
add esp, 8
```

```
mov edi, hline_buffer
mov ecx, [ebp + 16]
mov al, byte [ebp + 20]
cld
rep stosb
```

```
mov eax, 4
mov ebx, 1
mov ecx, hline_buffer
mov edx, [ebp + 16]
int 0x80
```

```
pop edi
pop ebx
leave
ret
```

```
; -----
; draw_vline(first_row, last_row, column, character)
; -----
```

```
draw_vline:
```

```
push ebp
mov ebp, esp
```

```

push ebx
push esi
push edi

mov esi, [ebp + 8]
mov edi, [ebp + 12]

.next_row:
    cmp esi, edi
    jg .done

    push dword [ebp + 16]
    push esi
    call goto_xy
    add esp, 8

    mov al, byte [ebp + 20]
    mov [one_char], al

    mov eax, 4
    mov ebx, 1
    mov ecx, one_char
    mov edx, 1
    int 0x80

    inc esi
    jmp .next_row

.done:
    pop edi
    pop esi
    pop ebx
    leave
    ret

; -----
; Macros
; -----

%macro PRINT_AT 4
    push dword %4
    push dword %3
    push dword %2

```

```
    push dword %1
    call print_at
    add esp, 16
%endmacro
```

```
%macro PRINT_CSTR_AT 3
    push dword %3
    push dword %2
    push dword %1
    call print_cstr_at
    add esp, 12
%endmacro
```

```
%macro PRINT_CSTR_N_AT 4
    push dword %4
    push dword %3
    push dword %2
    push dword %1
    call print_cstr_n_at
    add esp, 16
%endmacro
```

```
%macro HLINE 4
    push dword %4
    push dword %3
    push dword %2
    push dword %1
    call draw_hline
    add esp, 16
%endmacro
```

```
%macro VLINE 4
    push dword %4
    push dword %3
    push dword %2
    push dword %1
    call draw_vline
    add esp, 16
%endmacro
```

```
%macro ASK 6
    PRINT_AT %1, %2, %3, %4
    push dword %6
    push dword %5
```

```
    call read_input
    add esp, 8
%endmacro
```

```
;
=====
=====
; PROGRAM START
;
=====
=====
```

_start:

; Clear screen and show instructions.

```
mov eax, 4
mov ebx, 1
mov ecx, clear_screen
mov edx, clear_screen_len
int 0x80
```

PRINT_AT 1, 5, input_title, input_title_len

PRINT_AT 2, 5, input_instruction, input_instruction_len

; ABOUT ME questions

ASK 4, 5, p_name, p_name_len, name_buf, 79

ASK 6, 5, p_email, p_email_len, email_buf, 79

ASK 8, 5, p_course, p_course_len, course_buf, 79

ASK 10, 5, p_age, p_age_len, age_buf, 15

; MY FIRST questions

ASK 13, 5, p_happy, p_happy_len, happy_buf, 99

ASK 15, 5, p_achievement, p_achievement_len, achievement_buf, 99

ASK 17, 5, p_risk, p_risk_len, risk_buf, 99

; FAVORITES and HOBBIES questions

ASK 20, 5, p_colors, p_colors_len, colors_buf, 79

ASK 22, 5, p_song, p_song_len, song_buf, 79

ASK 24, 5, p_singer, p_singer_len, singer_buf, 79

ASK 26, 5, p_food, p_food_len, food_buf, 79

ASK 28, 5, p_weekend, p_weekend_len, weekend_buf, 99

ASK 30, 5, p_hobbies, p_hobbies_len, hobbies_buf, 99

```
; AMBITION and MOTTO questions
ASK 33, 5, p_ambition, p_ambition_len, ambition_buf, 99
ASK 35, 5, p_motto, p_motto_len, motto_buf, 99
```

```
; Wait before displaying final page
PRINT_AT 37, 5, ready_message, ready_message_len
```

```
push dword 7
push dword wait_buf
call read_input
add esp, 8
```

```
; -----
; Display completed slambook
; -----
mov eax, 4
mov ebx, 1
mov ecx, clear_screen
mov edx, clear_screen_len
int 0x80
```

```
; ABOUT ME ASCII header
PRINT_AT 1, 5, header1, header1_len
PRINT_AT 2, 5, header2, header2_len
PRINT_AT 3, 5, header3, header3_len
PRINT_AT 4, 5, header4, header4_len
PRINT_AT 5, 5, header5, header5_len
```

```
; Cat decoration
PRINT_AT 1, 64, cat1, cat1_len
PRINT_AT 2, 64, cat2, cat2_len
PRINT_AT 3, 64, cat3, cat3_len
PRINT_AT 4, 64, cat4, cat4_len
PRINT_AT 6, 14, divider_art, divider_art_len
```

```
; ABOUT ME box
HLINE 7, 1, PAGE_RIGHT, '-'
HLINE 13, 1, PAGE_RIGHT, '-'
VLINE 8, 12, 1, '|'
VLINE 8, 12, PAGE_RIGHT, '|'
```

```
PRINT_AT 7, 1, box_corner, box_corner_len
PRINT_AT 7, PAGE_RIGHT, box_corner, box_corner_len
PRINT_AT 13, 1, box_corner, box_corner_len
```

PRINT_AT 13, PAGE_RIGHT, box_corner, box_corner_len

PRINT_AT 8, 4, about_title, about_title_len

PRINT_AT 9, 4, name_label, name_label_len

PRINT_CSTR_N_AT 9, 12, name_buf, 30

PRINT_AT 10, 4, email_label, email_label_len

PRINT_CSTR_N_AT 10, 20, email_buf, 35

PRINT_AT 11, 4, course_label, course_label_len

PRINT_CSTR_N_AT 11, 13, course_buf, 25

PRINT_AT 12, 4, age_label, age_label_len

PRINT_CSTR_N_AT 12, 10, age_buf, 3

PRINT_AT 10, 48, tiny_heart, tiny_heart_len

; MY FIRSTS box

HLINE 15, 1, PAGE_RIGHT, '-'

HLINE 20, 1, PAGE_RIGHT, '-'

VLIN 16, 19, 1, '|'

VLIN 16, 19, PAGE_RIGHT, '|'

PRINT_AT 15, 1, box_corner, box_corner_len

PRINT_AT 15, PAGE_RIGHT, box_corner, box_corner_len

PRINT_AT 20, 1, box_corner, box_corner_len

PRINT_AT 20, PAGE_RIGHT, box_corner, box_corner_len

PRINT_AT 16, 4, firsts_title, firsts_title_len

PRINT_AT 17, 4, achievement_label, achievement_label_len

PRINT_CSTR_N_AT 17, 30, achievement_buf, 38

PRINT_AT 18, 4, risk_label, risk_label_len

PRINT_CSTR_N_AT 18, 29, risk_buf, 39

PRINT_AT 19, 4, happy_label, happy_label_len

PRINT_CSTR_N_AT 19, 43, happy_buf, 16

PRINT_AT 17, 61, flower1, flower1_len

PRINT_AT 18, 61, flower2, flower2_len

PRINT_AT 19, 61, flower3, flower3_len

; FAVORITES and HOBBIES boxes

HLINE 22, 1, HALF_BOX_WIDTH, '-'
HLINE 22, RIGHT_BOX_LEFT, HALF_BOX_WIDTH, '-'
HLINE 29, 1, HALF_BOX_WIDTH, '-'
HLINE 29, RIGHT_BOX_LEFT, HALF_BOX_WIDTH, '-'

VLINE 23, 28, 1, '|'
VLINE 23, 28, LEFT_BOX_RIGHT, '|'
VLINE 23, 28, RIGHT_BOX_LEFT, '|'
VLINE 23, 28, PAGE_RIGHT, '|'

PRINT_AT 22, 1, box_corner, box_corner_len
PRINT_AT 22, LEFT_BOX_RIGHT, box_junction, box_junction_len
PRINT_AT 22, RIGHT_BOX_LEFT, box_junction, box_junction_len
PRINT_AT 22, PAGE_RIGHT, box_corner, box_corner_len

PRINT_AT 29, 1, box_corner, box_corner_len
PRINT_AT 29, LEFT_BOX_RIGHT, box_junction, box_junction_len
PRINT_AT 29, RIGHT_BOX_LEFT, box_junction, box_junction_len
PRINT_AT 29, PAGE_RIGHT, box_corner, box_corner_len

PRINT_AT 23, 4, favorites_title, favorites_title_len
PRINT_AT 23, 39, hobbies_label, hobbies_label_len

PRINT_AT 24, 3, colors_label, colors_label_len
PRINT_CSTR_N_AT 24, 12, colors_buf, 22

PRINT_AT 25, 3, song_label, song_label_len
PRINT_CSTR_N_AT 25, 10, song_buf, 25

PRINT_AT 26, 3, singer_label, singer_label_len
PRINT_CSTR_N_AT 26, 12, singer_buf, 22

PRINT_AT 27, 3, food_label, food_label_len
PRINT_CSTR_N_AT 27, 10, food_buf, 25

PRINT_AT 24, 39, weekend_label, weekend_label_len
PRINT_CSTR_N_AT 24, 57, weekend_buf, 13

PRINT_AT 25, 39, hobbies_label, hobbies_label_len
PRINT_CSTR_N_AT 25, 48, hobbies_buf, 21

PRINT_AT 27, 51, heart1, heart1_len
PRINT_AT 28, 50, heart2, heart2_len
PRINT_AT 28, 64, cd_art, cd_art_len

```
; AMBITION and MOTTO box
HLINE 31, 1, PAGE_RIGHT, '-'
HLINE 36, 1, PAGE_RIGHT, '-'
VLINE 32, 35, 1, '|'
VLINE 32, 35, PAGE_RIGHT, '|'
```

```
PRINT_AT 31, 1, box_corner, box_corner_len
PRINT_AT 31, PAGE_RIGHT, box_corner, box_corner_len
PRINT_AT 36, 1, box_corner, box_corner_len
PRINT_AT 36, PAGE_RIGHT, box_corner, box_corner_len
```

```
PRINT_AT 32, 4, ambition_title, ambition_title_len
```

```
PRINT_AT 33, 4, ambition_label, ambition_label_len
PRINT_CSTR_N_AT 33, 17, ambition_buf, 45
```

```
PRINT_AT 34, 4, motto_label, motto_label_len
PRINT_CSTR_N_AT 34, 12, motto_buf, 45
```

```
PRINT_AT 32, 60, pencil_art, pencil_art_len
```

```
; Footer
PRINT_AT 38, 5, footer, footer_len
PRINT_AT 39, 1, newline, 1
```

```
; Exit
mov eax, 1
xor ebx, ebx
int 0x80
```

```

AAAAA BBBB OOO U U TTTT M M EEEEE /\_\/\
 A A B B O O U U T MM MM E      ( o o )
AAAAA BBBB O O U U T M M M EEEE > ^ <
 A A B B O O U U T M M E /|_|\
 A A BBBB OOO UUUU T M M EEEEE

~**~**~**~**~**~**~**~**~**~**~**~**~

#-----#
[ ABOUT ME ]
Name: Zack Gabriel J. Valenzuela
School Email: zjvalenzuela@fit.edu.ph @@
Course: Computer Science
Age: 20
#-----#

#-----#
MY FIRSTS
First big achievement: Living .-.-
First risk I ever took: Moving forward (o o)
First time I felt completely happy: Completing the c |=|
#-----#

#-----@@-----#
FAVORITES Hobbies:
Colors: Blue Weekend: Sleeping
Song: Die for you Hobbies: Sleeping and cycling
 Singer: Joji
Food: Porkchop
                                     /\_\/\
                                    ( o o ) C---D
#-----@@-----#

#-----#
AMBIITION /===>
Ambition: N/A
Motto: Keep moving forward
#-----#

***** MY SLAMBOOK *****
```